

12 | OVERVIEW



Teaching Climate and Health: Preparing the Next Generation of Medical Professionals

Abstract

Doctors and nurses are often the first to see how climate change harms people, yet few have been trained to recognize it. Harvard Chan C-CHANGE develops and evaluates curricula that prepare health professionals to identify and address climate-related health impacts. This lesson explores the first residency curriculum on climate and health, community organizing training for clinicians, and why climate knowledge is now part of being a good doctor.

Introduction

Every day, health professionals navigate the front lines of climate change. They may see tropical diseases appearing in cooler northern places or patients suffering from extreme heat. Climate change increases risks of heat-related illness, infections, asthma, mental health disorders, and poor pregnancy outcomes. It also disrupts care itself: when Hurricane Maria destroyed an IV bag factory in Puerto Rico, doctors had to ration IV fluids. The American Medical Association supported adding climate change to medical education in 2019, and more than 70 health organizations have declared climate change a health emergency. Harvard Chan C-CHANGE is working to make sure future clinicians are ready.

How Do Scientists Study This?

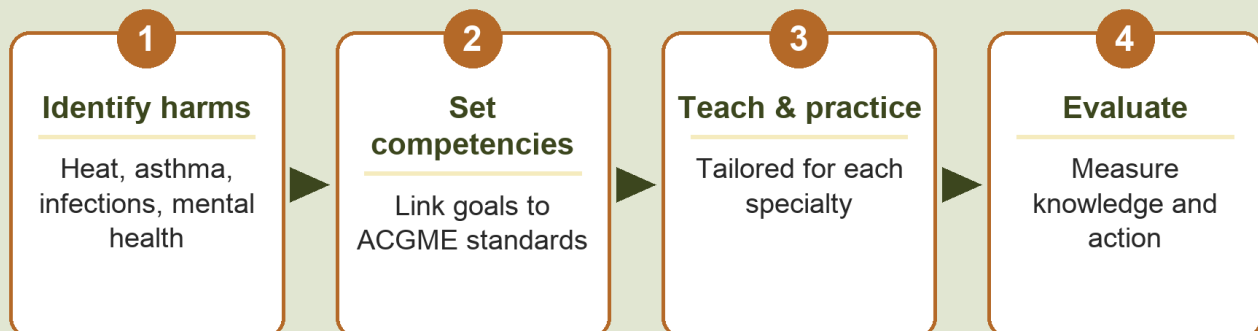


FROM THE ARTICLE

Students in a lecture at Harvard. Image: Harvard T.H. Chan School of Public Health.

C-CHANGE researchers design curricula and then evaluate whether they work. For medical residents, physicians from six medical schools created the first curriculum framework for teaching how climate change affects health, clinical care, and health care delivery. They linked each learning objective, learning format, and assessment strategy to the core competencies of the Accreditation Council for Graduate Medical Education (ACGME), which accredits residency programs. At Harvard Medical School, climate change was woven into all four years of the curriculum and its effects on students' knowledge were measured. A separate fellowship teaches health professionals community organizing skills, and its impact was also studied and published.

Building a climate and health curriculum



Study the Process

Define competencies

Learning goals were linked to the national standards that all residency programs must meet.

Teach across specialties

Content was tailored to emergency medicine, primary care, pediatrics, and mental health care.

Evaluate results

Researchers measured whether trainees' knowledge and capacity for climate action improved.

What Did They Discover?

The residency framework, published in Academic Medicine in 2020, was the first peer-reviewed resource to guide residency programs in teaching climate and health. It shows how each specialty needs different skills: emergency physicians need disaster response training, while primary care doctors and pediatricians can advise patients on how medications and outdoor activities raise heat illness risk. At Harvard Medical School, the new integrated curriculum significantly improved students' knowledge of climate-related health and equity topics. Training health professionals in community organizing significantly increased their capacity to take climate action. As Dr. Gaurab Basu put it, understanding how diseases change with climate helps doctors diagnose patients and plan treatment.

Climate in medical education

2019

AMA backs climate change in medical education

70+

health organizations call climate a health emergency

1st

residency climate curriculum framework, published 2020

Pediatrician Rebecca Philipsborn noted that in Atlanta, more extremely hot, polluted days are sending more children to emergency care for asthma attacks.

12 | TAKEAWAYS

Takeaways

Health is climate

Climate change worsens illnesses and disrupts supplies, so doctors must understand it.

Every specialty

Emergency, primary care, pediatric, and mental health clinicians each need tailored training.

Evaluate teaching

Testing curricula shows that climate training improves knowledge and action.

Key Terms

Residency

Hands-on training that new doctors complete in a medical specialty after graduating from medical school.

Curriculum framework

An organized plan of learning goals, teaching methods, and assessments for a course of study.

ACGME

The Accreditation Council for Graduate Medical Education, which sets standards for residency programs in the United States.

Competency

A skill or area of knowledge that a professional is expected to master.

Community organizing

Bringing people together to build shared power and take collective action on an issue.

12 | ACTIVITY

Activity

Specialty Climate Briefing

Students prepare a short briefing on how climate change affects patients in one medical specialty.

Materials: Specialty cards: emergency medicine, pediatrics, family medicine, psychiatry; Lesson text; Poster paper

1. Each group draws a specialty card.
2. List two climate-related health threats that specialty is likely to see, using examples from the lesson.
3. Write one skill a resident in that specialty should learn, and how you would test whether they learned it.
4. Add one way climate change could disrupt care in that specialty, such as supply shortages or power outages.
5. Present your briefing and compare how needs differ across specialties.

References

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